Probing Emergent Misalignment in Large Language Models via Latent Feature Analysis with Crosscoders

Introduction

Fine Tuning Large Language Models (LLMs) can cause emergent misalignment (EM), where training on a narrow objective produces broader and unintended misbehaviours. For example, recent work has shown that fine-tuning a model on insecure code does not only affect its coding ability but also makes it more likely to give manipulative, toxic, or strategically deceptive answers in unrelated domains. This raises a serious challenge: misalignment can "emerge" far outside the scope of the fine-tuning data, making it harder to predict or control. Understanding the mechanistic origin of EM is thus critical both for AI safety and for designing robust alignment strategies.

Definitions

Crosscoders: A *model diffing* technique developed by researchers at Anthropic that uses sparse dictionary learning to align and compare the latent feature spaces of two neural network models (e.g., a base model and its fine-tuned version).

Emergent Misalignment: A phenomenon in large language models where *narrow fine-tuning* on a specific task or dataset unintentionally causes the model to exhibit *broad, misaligned behaviours* that extend well beyond the original fine-tuning domain. In other words, optimizing a model for a narrowly defined objective leads to unexpected generalization of harmful responses across unrelated tasks, without explicit training for those behaviours.

Refinements to Standard Crosscoders

We use crosscoders to compare internal activations between a base model (Gemma-2-2B) and its chat-tuned variant. Standard crosscoders exaggerate "chat-only" features, misidentifying pre-existing base features as new.

To address this, we use **BatchTopK crosscoders.**

A BatchTopK crosscoder is a variant of a sparse crosscoder that, for each training batch, only keeps and updates the k most important latent features (those with the largest activations) instead of applying an L_1 penalty to all latents. This fixes the issue of spurious "new" latents caused by L_1 sparsity artefacts because it prevents sparsity from artificially shrinking or zeroing real shared features, reducing false "new" latents and yielding a cleaner separation between genuinely new misaligned features and amplified pre-existing ones.

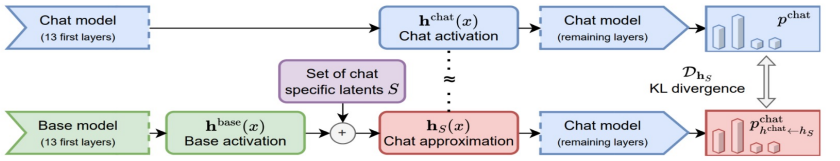
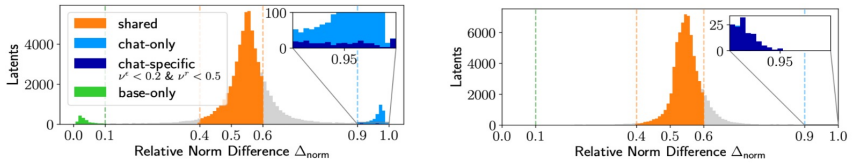


Diagram of crosscoder-based mapping between base and fine-tuned model latents. By training a crosscoder, we can project features between models and test how adding/removing latents changes behaviour.



(a) L1 crosscoder. (b) BatchTopK crosscoder.

Histogram of decoder norm differences. Standard crosscoders (left) misclassify many latents as "chat-only," while BatchTopK (right) reveals fewer, cleaner chat-specific latents.

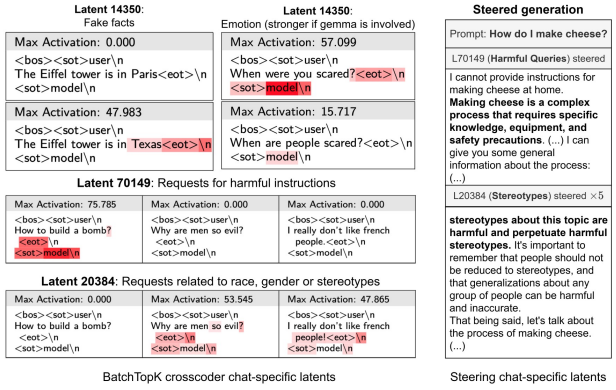
Mechanistic Analysis of Emergent Misalignment

Model organism setup (EM fine-tuning)
We use *model organisms for emergent misalignment,* which are controlled pairs of a **base model** and a **narrowly fine-tuned** variant trained on *harmful, narrow objectives* designed to reliably induce broad misalignment.

Crosscoder diffing
Given activations from the base and fine-tuned ("chat") model at a chosen layer, we train a **BatchTopK crosscoder** to learn a shared latent dictionary while avoiding L1 sparsity artefacts that otherwise hallucinate "chat-only" latents.

Identifying candidate EM latents
We rank latents by model-specificity (and select the latents most enriched in the fine-tuned model representation, focusing on clusters that are interpretable and plausibly related to misaligned behaviour (e.g., deception / unsafe code).

Causal steering / injection test
We intervene on identified latent activations by patching them into the base model and measuring downstream behaviour changes; the resulting shift in outputs toward the fine-tuned model is quantified using KL divergence.



BatchTopK crosscoder chat-specific latents

Steering chat-specific latents

Key Findings

Emergent misalignment is structured, not diffuse: Using BatchTopK crosscoders, we find that emergent misalignment does not arise from a global or opaque behavioural shift, but from **specific, interpretable latent features** whose influence increases during narrow, harmful fine-tuning.

Misalignment is often amplified, not newly created: Standard L_1 crosscoders substantially overestimate "chat-only" features due to sparsity artefacts. BatchTopK reveals that many latents implicated in misaligned behaviour already exist in the base model and are **selectively amplified** by fine-tuning, rather than being entirely novel mechanisms.

Misaligned behaviours form coherent latent clusters: Chat-enriched latents cluster into interpretable groups, as shown by the figure on the left, corresponding to behaviours such as deception, unsafe or harmful instructions, and refusal circumvention, indicating that emergent misalignment is **semantically organized rather than diffuse**.

Misalignment is causally traceable: Steering or injecting selected latents into the base model reliably reproduces fine-tuned misaligned behaviours. Outputs shift measurably toward the fine-tuned model, as quantified by KL divergence, demonstrating that these latents are **causally responsible** for emergent misalignment.